

NAME: _____ DATE: _____

6.9-1 Determining Molecular Formulas

1. Determine the molecular formula for compounds with the following empirical formulas and molar masses.

(a) CH_2 ; 84.18 g/mol

(b) ClCH_2 ; 98.96 g/mol

(c) HgCl ; 472.08 g/mol

(d) CH ; 78.12 g/mol

2. Combustion analysis reveals that a compound is 73.13 % carbon, 7.38 % hydrogen, with the remainder being oxygen. Mass spectrometry shows that the compound has a molar mass of 164.22 g/mol. Determine the empirical and molecular formulas for this compound.

3. Dimethylbenzenes, or xylenes, are hydrocarbons composed of two CH_3 - groups bonded to a 6-carbon ring. They have the same molecular formula, which is C_8H_{10} , and differ only in the placement of the CH_3 - groups on the ring. These compounds are flammable and sometimes used as accelerants in arson fires. A sample taken from the scene of a suspected arson fire is found to be 90.49 % carbon and 9.51 % hydrogen.

(a) Based on the percent composition information, is it possible that the sample is a dimethylbenzene? Explain your answer.

(b) Ethylbenzene's molar mass and percent composition is the same as that of dimethylbenzene. Ethylbenzene has one $\text{CH}_3\text{-CH}_2$ -group bonded to a 6-carbon ring rather than two CH_3 -groups bonded to a 6-carbon ring, as in dimethylbenzene. Would investigators be able to conclusively prove that the accelerant used was dimethylbenzene and not ethylbenzene? Explain.
